Intergraph Smart 3D Hangers and Supports SmartParts Configuration

Hexagon Documentation

Generated 04/02/2026

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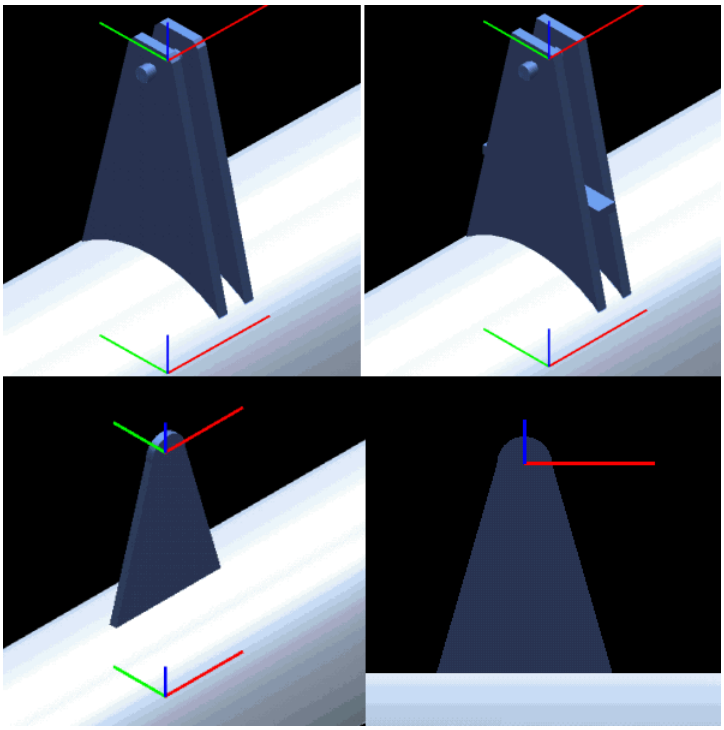
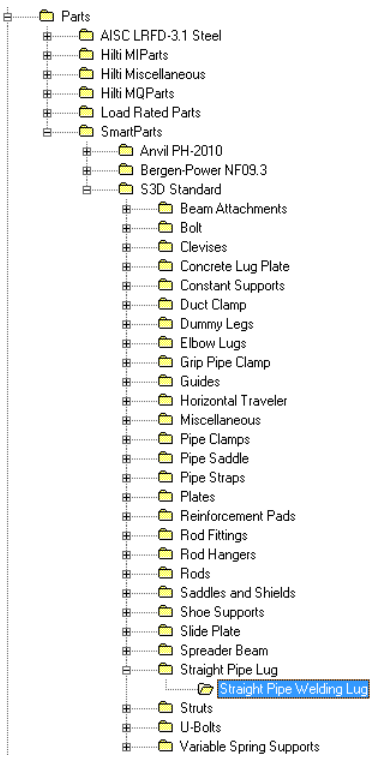
Straight Pipe Lug

Library Name: SmartParts
Workbook: HS_S3DParts.xls
Codelist: HS_S3DParts_Codelist.xls
ProgID: HSSmartPart,Ingr.SP3D.Content.Support.Symbols.PipeLug
Part Name: Straight Pipe Welding Lug
Part Description: Single/Double Lug With Curved top/Square top, Parallel
Single/Double Lug With Curved top/Square top, Perpendicular
Double Lug With Stiffener/Rounded top/Chamfered top



To place a sample Straight Pipe Welding Lug SmartPart delivered with the catalog:

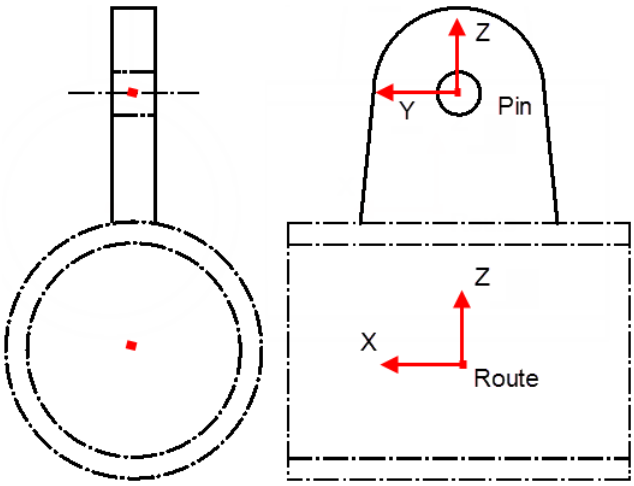
1. Click **Tasks** > **Hanger and Supports**.
2. Place a design support on a pipe.
3. Start the **Place Part** command, and then select the design support.
4. In the **Select Part** dialog, select **Parts > SmartParts > S3D Standard > Straight Pipe Lug > Straight Pipe Welding Lug** as shown below.



Part Properties

Millimeters (mm) is the default unit. Use "in" at the end of dimensional values to specify inches. Ports are shown as points highlighted in red.

Local Coordinate System



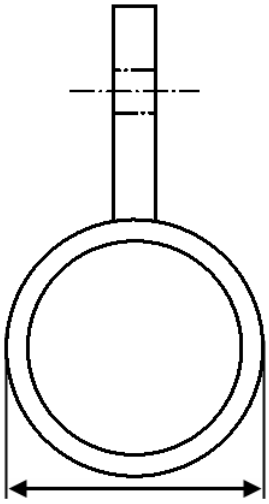
Basic Straight Pipe Lug Attributes

IJUAhsRodDiameter::RodDiameter

Specifies the rod diameter that is used by the PipeLug and the part selection rule.

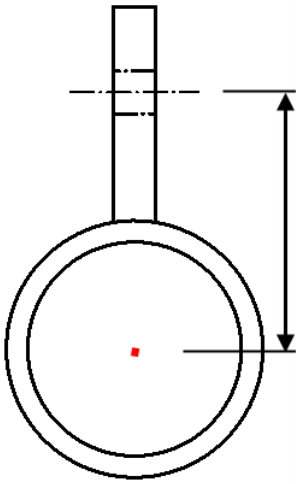
IJOAhsPipeOD::PipeOD

Specifies the outer diameter of the pipe that is used with the PipeLug.



IJUAhsRodTakeOut::RodTakeOut

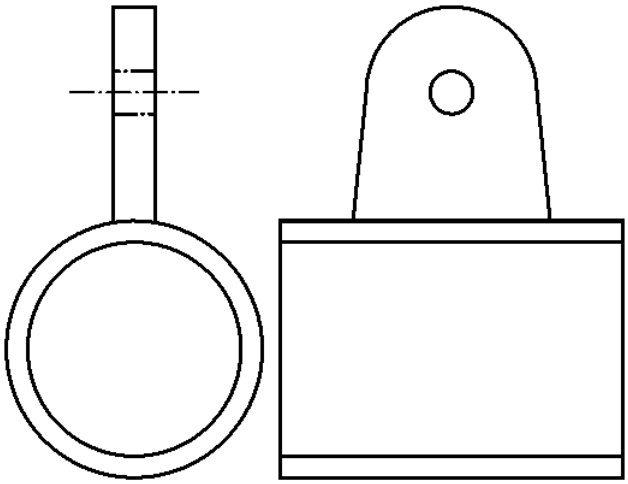
Specifies the distance between the horizontal pipe center line and the top lug port.



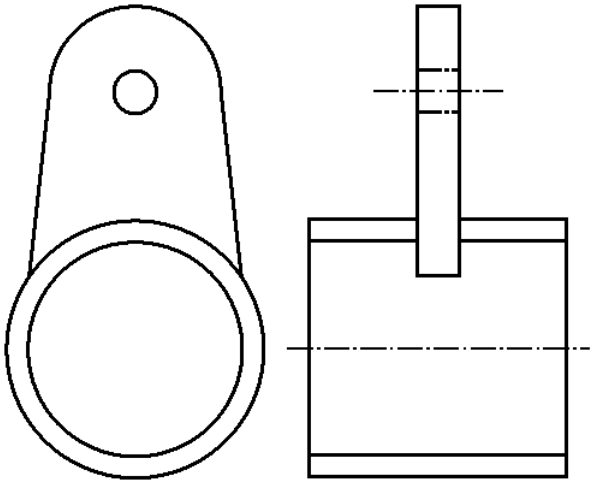
IJUAhsIsPerpendicular::IsPerpendicular

Specifies whether the lugs are parallel or perpendicular to the pipe axis. This attribute accepts a boolean value.

IsPerpendicular - 0 (False)



IsParallel - 1 (True)

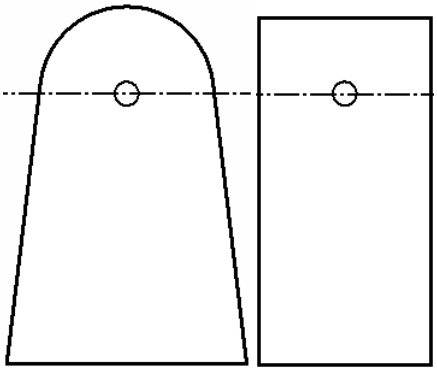


Side Attributes

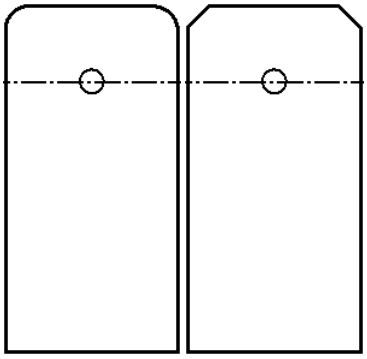
IJUAhsLug::TopShape

Specifies the shape that is used for the top of the lug.

1 - Curved 2 - Squared

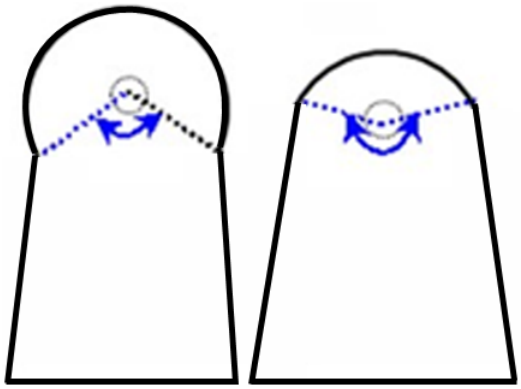


3 - Rounded 4 - Chamfered



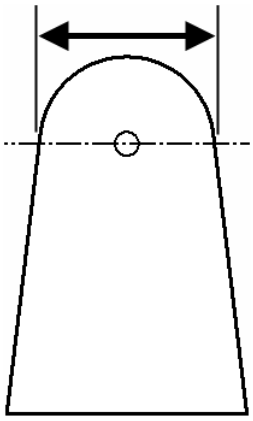
IJUAhsAngle1::Angle1

Specifies the angle between the start and the end points of the top curve. If the **TopShape** is not curved, then the software ignores this attribute. The Angle1 value must be positive and greater than 90 degrees and less than 270 degrees, else the software displays a warning message and resets the **Angle1** value to 180 degrees.



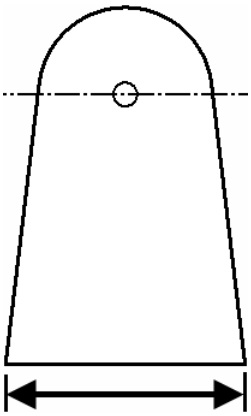
IJUAhsWidth1::Width1

Specifies top width of the pipe lug. If **Width1** is set to zero, then **Width2** is used.



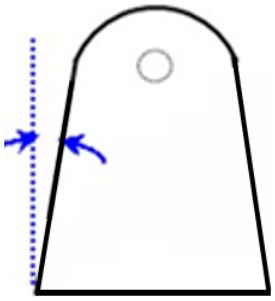
IJUAhsWidth2::Width2

Specifies the bottom width of the pipe lug. If the **Width2** and **Angle3** values are zero, then **Width1** is used instead.



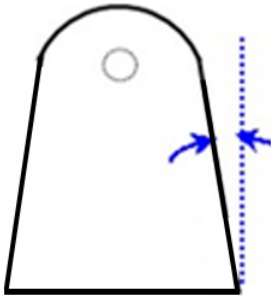
IJUAhsAngle2::Angle2

Specifies the angle of the left tapered portion. If you do not assign a positive value to **Angle2** that is less than 30 degrees, the software displays an error message. If **Angle2** is set to zero, then the left line is drawn as global vertical.



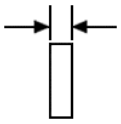
IJUAhsAngle3::Angle3

Specifies the angle of the right tapered portion. If you do not assign a positive value to **Angle3** that is less than 30 degrees, the software displays an error message. **Angle3** overrides **Width2** to get the tapered bottom point.



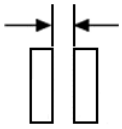
IJUAhsThickness1::Thickness1

Specifies the thickness of the lug.



IJUAhsGap1::Gap1

Specifies the gap between the lug components of a double lug. If a value is not specified, then a single lug is drawn.



IJUAhsLug::StiffenerOffset

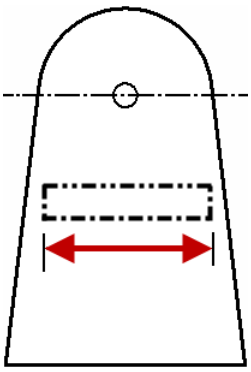
Specifies the distance a stiffener is offset between the two components of a double lug. **StiffenerOffset** is set from the center of pin port.

IJUAhsLug::StiffenerHeight

Specifies the height of the stiffener between the two double lug components.

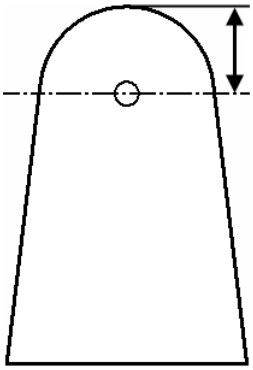
IJUAhsLug::StiffenerLength

Specifies the length of the stiffener between the two double lug components.



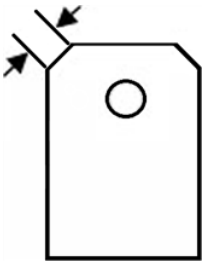
IJUAhsOffset1::Offset1

Specifies the distance between the top of the lug and the top port. When **TopShape** is set to **1 - Curved**, **Offset1** is used as the eye radius.



IJUAhsLug::ChamfLength

Specifies the length of the chamfered edges. This attribute is valid only when **TopShape** is set to **4 - Chamfered**.



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